

Skew-Symmetric Error Distribution Analysis of Elasticity Dynamics of Firm Financial Statement Components: A Case Study in Indonesian Stock Markets using The Cobb-Douglas Linearized Log Form

Galvin Adriel Darvell Nadapdap
Accounting Department
Bina Nusantara University
Jakarta, Indonesia
galvin.nadapdap@binus.ac.id

Alexander Benedick
Accounting Department
Bina Nusantara University
Jakarta, Indonesia
alexander.benedick@binus.ac.id

Sasya Sabrina, S.E., M.M., CPSAK
Accounting Department
Bina Nusantara University
Jakarta, Indonesia
sasyasabrina@binus.ac.id

Abstract—The main purpose of this paper is to explore the impact of total assets and number of employees to the dynamics of total revenue via the Cobb-Douglas production function log-linear regression form adapted into Skew-Symmetric form via Broyden–Fletcher–Goldfarb–Shanno algorithm (BFGS), of Indonesian firms listed on the Indonesian Stock Exchange (IDX). This study is important since it not only explores how firms' revenue change by its balance sheet inputs on the Indonesian stock market but also explores the Skew-Symmetric modelling applied to regression modelling. The results show a very strong positive correlation of total assets and number of employees to firm revenue. Specifically, an increase of one percentage point in total assets is associated with an increase of 0.41865 in total revenue ($p < 0.001$) and an increase in one percentage point in number of employees is associated with an increase of 0.21937 in total revenue ($p < 0.001$). The findings suggest that as firms accumulate more input via the reconciliation of total assets and number of employees as capital and labor respectively, outputs in form of total revenue also accumulate proportionally. For future research, it is recommended to use larger sample sizes and longer observation periods to confirm these findings and explore additional variables.

Keywords— Skew-Symmetric, Cobb-Douglas production function, Brayden-Fletcher-Goldfarb-Shanno (BFGS) Algorithm

I. INTRODUCTION

The statistical modelling of financial statements has been a primary frontier upon the analysis of financial statements and in recent years, advanced econometric modelling has gained significant traction, reflecting a deeper emphasis on capturing the complexities of resource utilization and economic growth. Readings such as Chris Brooks, use an econometric approach to study UK house prices via time-series models and rental values using principal components analysis (PCA) [1], and also Damodoran uses a traditional regression model in financial statements analysis for valuating stock values via financial components such as debt-to-equity ratios, using traditional linear regression coefficients [2].

The primary factor behind the significant contribution of statistical analysis to the analysis of financial statements is due to its enhanced accuracy and predictive power, by integrating concepts and practices from big data analytics that facilitates applications from fraud to risk analysis. Studies, for example from Coenders et.al., show that integrating compositional data analysis into financial statements enhances the understanding of the financial data itself [3].

In response to the advances in statistical modelling on the discipline of finance and accounting, environments of the capital market are one of the playing fields as the focus of analysis, for which it measures an aggregate of economic variables from firms via its equity, capital, labor, and much more. The Indonesian capital market, with its rapid growth

and diverse industries, provides a unique environment for studying the dynamics of financial performance, particularly through the lens of firm economic variable elasticity. One study by Bank Indonesia in a financial stability review, shows that Indonesian firms operate in an emerging market characterized by economic volatility and rapid development in emerging markets especially during the economic recovery of COVID-19 [4]. The environment of this rapid development offers an opportunity to analyze Total Factor Productivity (TFP) of the Indonesian market using classic economic models such as the Cobb-Douglas production function that is interchangeable from a mathematical form to a linear regression form. One study by Solow in post US WW2 economy, uses the production function model to study the aggregate production by analyzing firm's labor and capital via an econometric regression [5]. Similar studies of the Indonesian market use this method incorporating a stochastic frontier approach to incorporate the model with volatilities that leads to an economic crisis in the market such as the 1997 crisis [6].

The reconciliation between the firm's labor and capital in the books is reflected mainly through its total assets and number and employees, by which the Cobb-Douglas production function provides. The Cobb-Douglas production function is one of the main models used in economic analysis from microeconomics to macroeconomics. The usage of the Cobb-Douglas production function model in modelling TFP implies the modelling of variables into a traditional linear regression model, by which the model uses a logarithm form to measure its variables. Traditional regression models often rely on the assumption of normally distributed residuals, which can be problematic when dealing with skewed data or outliers. Studies by Azzalini confirm the limitation of classical likelihood functions that are inherent to linear regression models, limiting the analysis to not incorporate skewness and fat heavy tails [7]. To address this limitation, this study proposes the use of skew-symmetric error distribution analysis within the Cobb-Douglas framework. This approach allows for more accurate estimations of elasticity dynamics in financial statement components, particularly in the context of Indonesian firms, where traditional assumptions may not hold.

Many of prior research uses the Cobb-Douglas production function to study the reconciliation of inputs to output of production of a firm. The work of Sarkar et.al, uses the Cobb-Douglas production function with constrained optimization to model revenue through the inputs and outputs of IT infrastructure via new server costs and power and cooling costs with conclusion of significance and good to fit an $R^2 = 0.99$ [8]. Another study conducted by Jimichi et.al, uses the variables of revenue, total assets, and number of employees

that incorporated a skew symmetric approach to the Japanese capital market to analyze trends in revenue sectors of industries [9].

However, these studies focus on contexts outside Indonesia and do not fully detail the optimization process for coefficient estimation. To the best of our knowledge, no study has applied a skew-symmetric Cobb–Douglas framework to Indonesian listed firms. This is important because the Indonesian market has unique institutional characteristics—such as high state ownership, resource dependence, and growing digital sectors—that may alter production elasticities. Our contribution is therefore twofold, where first is adapting and optimizing the Cobb–Douglas and skew-symmetric models to Indonesian firm-level quarterly data, and second demonstrating the feasibility of coefficient estimation under data constraints that are specific to emerging markets.

II. LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

A. Literature Review

In the research, we integrate two theories. The first theory is the Theory of Production. According to Cobb and Douglas, the firm combines and reconciliates the inputs of labor and capital into production. Labor and capital have their trade-offs between one and another in respective elasticities, by which one can utilize or prefer one or the other [10]. In financial statement analysis, the Cobb–Douglas function serves as a framework to assess how efficiently a firm allocates resources. Empirical studies have validated this function in corporate finance, showing that asset utilization and workforce management significantly impact firm revenue, such as the study of Hall & Jones [11].

The second theory that we utilize is the Theory of the Firm. According to Coase, firms exist to minimize transactions costs and operate efficiently in response to market forces and aim to maximize profits subject to competitive pressures [12]. In financial statement analysis, this theory underpins key financial ratios, such as return on assets (ROA), return on equity (ROE), and revenue efficiency metrics, to evaluate firm performance [13].

Combining both of these theories is essential in investigating and quantifying how the firm utilizes its resources for maximizing the output of the firm to bring profits and respond to market forces, in which both theories can be formulized using the Cobb–Douglas production function to explain the firm's movement in the market.

B. Effects of Total Assets, Total Employees on Total Revenue in the Indonesian Market

Total assets, total employees, and revenue are fundamental indicators of a firm's financial health and operational capacity. The Cobb–Douglas production function has been widely utilized in economics to model the relationship between these variables, emphasizing how firms allocate resources to maximize output. This function assumes that firm output is a function of inputs such as capital (proxied by total assets) and labor (proxied by the number of employees), with revenue serving as the dependent variable. From an economic perspective, total assets represent the firm's ability to invest in productive capacity, including physical infrastructure, technology, and working capital. A higher asset base is expected to contribute positively to revenue generation. Similarly, the number of employees reflects the human capital available to a firm, with more employees generally leading to

increased productivity and output. However, the relationship is subject to diminishing marginal returns, meaning that after a certain point, additional assets or employees may not significantly enhance revenue.

According to classical production theory, firms strive to balance these inputs to optimize performance. Previous research indicates that firms with well-allocated assets and efficient labor management exhibit stronger revenue growth. Thus, based on economic logic and financial theory, we propose the following hypothesis:

H_1 : Variables of Total Assets and Total Employees correlate significantly to a firm's Total Revenue.

We also note that we intend to capture “pure output” and “pure inputs” into our study, such that we do not incorporate variables such as expenses, net profit, and return ratios such as Return on Assets (ROA), and Return on Equity (ROE). The purpose of this is to isolate direct production and operational capacity of firms, consistent with the core objective to the paper.

C. Comparison to the Skew-Symmetric Distribution

Traditional econometric models, such as Ordinary Least Squares (OLS), assume that error terms follow a symmetric normal distribution. However, financial data often exhibit skewness due to market inefficiencies, investor behavior, or firm-specific shocks. Skew-symmetric distributions offer a more flexible modelling approach by allowing asymmetric error terms, which better capture real-world financial dynamics.

In the context of financial statement analysis, incorporating a skew-symmetric regression framework enhances the robustness of parameter estimation, reducing biases associated with mis-specified normality assumptions. For example, firms with highly volatile financial structures such as those experiencing rapid asset growth or workforce expansion may not conform to symmetric statistical assumptions. By integrating skew-symmetric distributions, we can more accurately model the elasticity of revenue concerning total assets and total employees.

Research in financial econometrics suggests that skew-symmetric models improve prediction accuracy and explanatory power, particularly in emerging markets where financial anomalies are more pronounced. By testing these models within Indonesian firms, we aim to provide empirical evidence supporting their application in financial statement analysis. Thus, based on statistical advancements and financial theory, we propose the following hypothesis:

H_2 : A regression model incorporating skew-symmetric distributions provides an improved estimation of firm performance compared to traditional symmetric regression models.measures proportionately more than is customary.

III. METHODOLOGY AND FORMULATION

A. Material and Method

The research conducted in this paper uses quantitative analysis by utilizing secondary data sources including total assets, revenue, number of employees and financial statement numbers obtained from Yahoo Finance imported using Python, with 89 sample companies listed on the Indonesian Stock Exchange (IDX) with multiple industries covered , with time interval between Q2-2023 and Q2-2025 (2 years). We

use 2-year dataset with quarterly intervals since the limitation of Yahoo Finance being unable to extract more than 1 year of financial statement components. The total observations of data are 575 observations. The data can be accessed through *V1_indonesian_company_quarterly_data_2023_2025.xlsx*. The R-language software is used to compute and run the algorithms, including the log-likelihood function and the primary regressions.

B. Statistical Formulation

The Cobb-Douglas production function has been widely applied in economics and finance to model firm productivity and financial performance. Traditionally, it is expressed as the form:

$$R = \kappa A^{\beta_1} N^{\beta_2} \epsilon \quad (1)$$

Where $\kappa, R_t, A_t, N_t, \epsilon_t$ denote constant parameters of regression, revenues, assets, and number of employees and an error term of regression at time t respectively. The general form originally proposed by Charles Cobb and Paul Douglas is $Q = AK^\alpha L^{1-\alpha}$ in which Q denotes the output. α is the elasticity parameter, L denotes labour, and K denotes capital. This function satisfies key economic properties, such as constant returns to scale and diminishing marginal returns which was studied by Solow and Coelli et al [14]. In financial modeling, the elasticity parameters β_1 and β_2 help assess the relative impact of assets and labor on firm performance, by which studies by Hall and Jorgensen [15] showed this. To formulate the regression, we take the logarithm of the equation, and we will get the following:

$$\log R = \beta_0 + \beta_1 \log A + \beta_2 \log N + \epsilon_{LN} ; \epsilon_{LN} \sim \mathcal{LN}(\mu = 0, \sigma^2 = 1) \quad (2)$$

Where $\mathcal{LN}$ denotes the Log-Normal distribution with mean 0 and variance 1. The form that is in this equation is a classical log-linear regression model with the assumption of the log-normal distribution. The economic assumption behind the model addresses the diminishing margin of production of a firm, based on the original model of Cobb's and Douglas' theory and formulation of firm production economics.

The classical econometric approach assumes normality in error terms, yet financial data often exhibit skewness and heavy tails. To address this, skew-symmetric distributions provide a more flexible alternative, capturing asymmetric risk structures in financial returns studied by Azzalini. These distributions modify classical likelihood functions, allowing for non-normal residuals in regression models, by which they studied by Arellano & Bover [16]. The skew symmetric distribution is defined as:

$$f(\epsilon) = \frac{2}{\omega} \phi\left(\frac{\epsilon - \xi}{\omega}\right) \Phi\left(\frac{\epsilon - \xi}{\omega}\right) \quad (3)$$

Where $\phi, \Phi, \epsilon, \omega, \xi$ is the standard normal probability density function, standard normal cumulative distribution function, location parameter, and the skewness parameter. The algorithm that is used by the statistical software will maximize the likelihood function and produces an output

such that it reformulates the linear regression error term to incorporate skewness. Thus (2) will be revised as:

$$\log R = \alpha_0 + \alpha_1 \log A + \alpha_2 \log N + (\epsilon \sim f(\epsilon)) \quad (4)$$

Using R-language, we optimize the parameters using the R-language built in optimization packages. One of the optimization methods that we will utilize is the Brayden-Fletcher-Goldfarb-Shanno (BFGS) method. The BFGS method is optimization conducted for functions or equations that have no constraints or in other words unconstrained optimization. We will use this optimization to obtain the new coefficients of (4) via the distribution formulated in (3). The formulation of the BFGS requires the formulation of the log-likelihood function of which:

$$\ell(\theta) = \sum_{i=1}^n \log \left[\frac{2}{\omega} \phi\left(\frac{\epsilon_i - \xi}{\omega}\right) \Phi\left(\alpha \cdot \frac{\epsilon_i - \xi}{\omega}\right) \right] \quad (5)$$

Where $\theta = (\alpha, \beta, \xi, \omega, \alpha_{\text{skew}})$ is the parameters for the log-likelihood of the skew-symmetric distribution. Thus, the BFGS algorithm is formulated by the relation of the parameter θ where at iteration $k + 1$:

$$\theta_{k+1} = \theta_k - H_k^{-1} \nabla \ell(\theta_k) \quad (6)$$

Where H_k is the Hessian matrix of second-order partial derivatives via the log-likelihood function. The update rule for the BFGS algorithm starts with the identity matrix of $H_0 = I$ with initial guess θ_0 . The complete methodology in R-language is summarized in the table below.

TABLE I. ALGORITHM TABLE

Step	Description/Mathematical Formulation
Data Import	Read dataset $\mathbf{D}$ Excel, convert date column into proper format
Variable Transformation	Compute $LR = \ln(\text{Revenue})$, $LA = \ln(\text{Assets})$, $LN = \ln(\text{Employees})$
Data Cleaning	Exclude data observations with $R, A, N \leq 0$. Or more formally: $\mathbf{D} \leftarrow \{x \in \mathbf{D} R, A, N > 0\}$
Classical Cobb-Douglas Model	Run $\log R = \beta_0 + \beta_1 \log A + \beta_2 \log N + \epsilon$
Prediction Grid	Generate LA_j, LN_k and predict $\widehat{LR}_{jk}$ from fitted model
Skew-Symmetric Model	Define : $\ell(\theta) = \sum_{i=1}^n \log f_{SN}(\epsilon_i \xi, \omega, \alpha)$ for the likelihood function
Estimation	$\hat{\theta} = \arg \min_{\theta} -\ell(\theta)$ via BFGS where $\theta = (\beta_0, \beta_1, \beta_2, \xi, \omega, \alpha)$ with starting parameters (0,1,1,0,1,0), respectively
Diagnostics	Residual plots, Shapiro-Wilk test, Breusch-Pagan test, and Durbin-Watson test
Model Comparison	Compare Classical vs Skew models using the AIC and BIC
Visualization	Plot observed vs predicted values, residual histograms, Q-Q plots, and 3D surface plots of predicted Log Revenue

IV. RESULTS AND DISCUSSION

A. Descriptive statistics, regressions, and optimization results

Descriptive statistics were conducted in order to understand the general outline of the data in terms of its

TABLE I. DESCRIPTIVE STATISTICS

Variable	Obs	Mean	SD	Median	Trimmed	MAD	Min	Max	Range	Skew	Kurtosis	Standard Error
Log Revenue	575	29	4.38	30.74	29.54	2.21	18.2	34.4	16.24	-1.06	-0.39	0.2
Log Assets	575	31.26	4.46	32.92	31.7	2.15	21.3	37.7	16.41	-0.98	-0.38	0.2
Log Number of Employees	575	10.7	1.43	10.61	10.71	1.44	6.17	14.1	7.95	-0.11	0.09	0.1

minimum, maximum, mean, standard deviation, skewness, and kurtosis. We first of all transformed the data into logarithmic form to perform the regression while also to shrink the base numbers smaller for practicality of analysis. The statistics are outlined in Table II.

The sample company average log revenue is 29, with a negatively skewed distribution of -0.39, indicating a longer left tail, thus a concentration of firms with higher revenues. The distribution is slightly platykurtic i.e. flatter than the normal distribution, suggesting lighter tails and fewer extreme values than its normal counterpart. Similarly, log assets show a negative skewness as well as log employees. However, for the log assets, this shows that many firms have large asset bases whilst the log normal is approximately symmetric since its skewness and kurtosis is closer zero implying that employee counts evenly distributed across firms. Thus, the data shows that the variables are heavily skewed.

Next, we regress the model using classical regression techniques without optimizing the parameter coefficients of via the skew symmetric distribution. The results are in Table II below. The classical regression yields a really good estimate of the independent variables, with a significance of coefficients of the total assets and total employees of under 0.001, meaning that the independent variables have a very significant influence on the firm's total revenue. The R-squared is 0.9343, indicating a significant overall fit of 93.43% of variability of the dependent variable (Log Revenue) to its independent variables. Thus, confirming our first hypothesis (H_1) that firm revenues significantly explain the firm's variations in revenue.

TABLE II. CLASSICAL REGRESSION RESULTS

Variable	Estimate	Std.error	t-value	Pr(> t)	R-square	0.9343
Constant	-2.05587	0.39748	-5.172	3.20e-07 ***	p-val	< 2.2e-16
Log Total Assets	0.90882	0.01213	74.899	< 2e-16 ***	F-stat	4064 on 2 and 572 df
Log Number of Employees	0.24789	0.03783	6.552	1.27e-10 ***		

p-val's : <*0.05, <**0.01, <***0.001

The next step is to regress the skew-symmetric model in order to compare both models in the later sections. The results are in Table III and the optimized parameters using the BFGS algorithm are displayed in Table IV. The visuals are also represented in Fig.1 and Fig.2. We can see that the scatterplot represents a really good fit for the function, and thus for the 3D graph of the production function yields as close resemblance to the original Cobb-Douglas production graph with the elasticities of each variable defined as the coefficient of regression. The surface rises smoothly in the graph as assets and employees increase, reflecting a positive marginal

product in both inputs whilst also showing diminishing returns via the curvature of the slope.

TABLE III. OPTIMIZATION RESULTS

Parameter	α_0	α_1	α_2	ζ	ω	α
Estimate	-0.5389	0.10564	0.9699	-0.5386	1.73185	-3.1546

TABLE IV. SKEW-SYMMETRIC REGRESSIONS

Variable	Estimate	Std.error	t-value	Pr(> t)	R-square	0.9992
Constant	-11.3699	3.467209	-3.279	0.001116 **	p-val	< 2.2e-16
Log Total Assets	1.280150	0.09737	0.088142	< 2e-16 ***	F-stat	5341 on 87 and 319 DF
Log Number of Employees	0.033857	0.10320	5.494	0.002572**		

p-val's : <*0.05, <**0.01, <***0.001

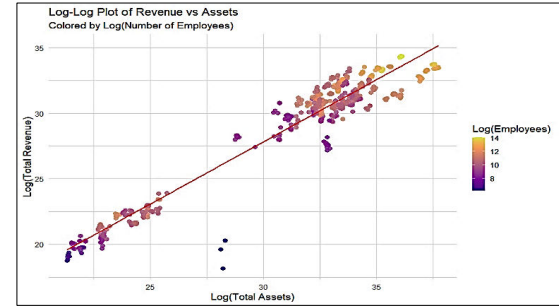


Fig. 1. The scatterplot of the log data with a trendline with gradient coloring. The data shows small deviations from the average to some large deviations from the trendline

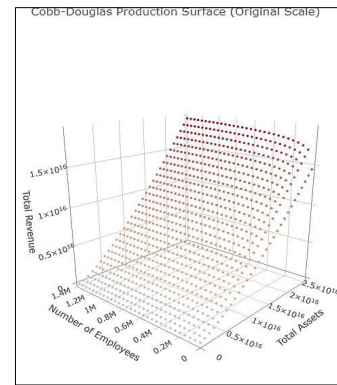


Fig. 2. The Cobb-Douglas production function plot from the empirical data. The plot resembles to the original Cobb-Douglas production function form graph

The Skew-Symmetric regression produced higher explanatory power ($R^2 = 0.9993$) but different coefficient magnitudes compared to the Classical model. For example, the Log Total Assets coefficient drops from 1.2801 in the

Classical regression to 0.0339 in the Skew-Symmetric, while the employee elasticity also shifts markedly. These differences highlight how model choice affects interpretation, though both support our first hypothesis (H_1). To evaluate our second hypothesis (H_2), we further compared models using normality, heteroscedasticity, autocorrelation, and information criteria tests.

B. Normality Tests

In this study, we performed the normality test using the Shapiro-Wilk test via R-language's normal probability plot and Shapiro-Wilk test function. Variables that passed the normality test of the Shapiro-Wilk will have a W-statistic closer to one. Given the original variables are heavily skewed, we proceed in testing the residuals of the model to determine which model satisfies the normality assumption. The statistical results are shown in Table V as well as in Fig.3 for the Q-Q plot comparison.

TABLE V. SHAPIRO-WILK NORMALITY TEST

Model	W Statistic	p-value
Classical	0.86591	< 2.2e-16
Skew Symmetric	0.98092	1.643e-06

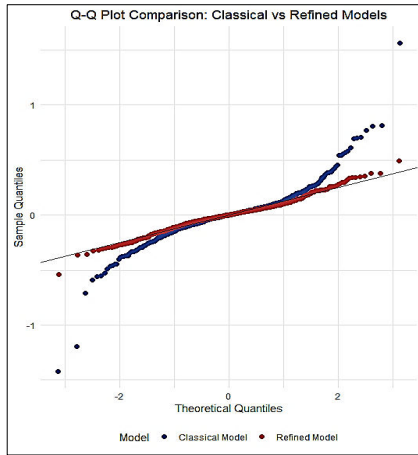


Fig. 3. Normal Probability Plot. The Skew-Symmetric Distribution is flatter and has less outliers than the Classical model, in which confirms the better estimate for the Skew-Symmetric model

The Skew-Symmetric model outperforms the Classical model in terms of residual behavior. Its Shapiro-Wilk statistic ($W = 0.98092$) is closer to one, indicating residuals more consistent with normality, even though both models formally reject strict normality at the 5% level. Q-Q plots reinforce this result: residuals from the Classical model deviate more from the 45° line and display heavier tails, while those from the Skew-Symmetric model align more closely with the reference, showing fewer outliers. Together, these diagnostics confirm that the Skew-Symmetric specification better addresses skewness and kurtosis, thereby improving model fit.

C. Autocorrelation and heteroskedasticity tests

The next assumption test is correlation and heteroskedasticity, in which we employ Breusch-Pagan test for heteroskedasticity and the Durbin-Watson for

autocorrelation, in which both are summarized in Table VI. The d-statistic is reported 2.2518 for the classical model and 2.1207 for Skew-Symmetric model. Typically, a d-statistic near 2 suggests no autocorrelation, while less than 2 suggests a positive autocorrelation. Based on the results, both models have no significant autocorrelation in the residuals of the regression model, since both are closer to 2. However, Skew-Symmetric Model has a value much closer to 2 with 0.0871 decimal points deviation. Thus, in terms of Autocorrelation, the Skew-Symmetric model is far better in autocorrelation terms.

Another test is the BP test, by which the Classical model shows a BP-statistic of 214.21 and the Skew-Symmetric model shows a BP-statistic 195.21. Typically, a lower p-value ($p < 0.05$) and a high BP-statistic indicates heteroskedasticity. Based on the results both models show heteroskedasticity. However, the BP-statistic for the Skew-Symmetric is far significantly lower than the classical model, the p-value is higher than the Classical model. Thus, the Skew-Symmetric model is still better for estimates than the Classical model although both show patterns of Heteroskedasticity.

TABLE VI. DURBIN-WATSON AND BRESUCH-PAGAN TESTS

Model	Breusch-Pagan Test			Durbin-Watson test
	BP Statistic	Degrees of Freedom	p-value	
Classical Model	214.12	88	4.201e-10	2.2518
Skew-Symmetric Model	195.21	89	2.592e-12	2.1207

D. Information Criterion Tests

The final evaluation uses information criteria (IC), which balance model fit against complexity to prevent overfitting. We report both the Akaike Information Criterion (AIC), focused on predictive accuracy, and the Bayesian Information Criterion (BIC), which incorporates a stronger penalty term. In both cases, lower values indicate a better model. Results are shown in Table VII.:

TABLE VII. INFORMATION CRITERION TESTS

Metric	Classical	Skew-Symmetric Model
AIC	25.96	-519.94
BIC	422.21	-133.70

The AIC and BIC in the Skew-Symmetric model are lower by a large margin in comparison to the Classical model, meaning that the likelihood of the Skew-Symmetric model is much higher both in predictability and fitting the data. Thus, from the tests we can conclude that the Skew-Symmetric regression provides improved fit diagnostics to the classical regression which assumes log-normal distributions upon the residuals, hence confirming the second hypothesis (H_2).

E. Discussion Comparison with existing studies and interpretation

Our findings are consistent with prior applications of Cobb-Douglas and skew-symmetric models, such as Jimichi

et al. in the Japanese capital market. Yet the Indonesian context differs in important ways. Japan has long exhibited high capital intensity and shifting capital and labor substitution patterns, shaped by demographics and corporate investment, which helps explain why asset elasticities in Japanese studies often diverge from those in emerging markets [17]. In contrast, Indonesia's declining manufacturing share and pivot toward commodities and resource sectors create a different firm composition and investment profile, raising aggregate capital intensity and cross-firm heterogeneity [18]. This is visible in Figure 2, where Indonesian revenue appears more strongly driven by asset growth, is consistent with a capital-intensive structure (with mining firms as a primary example).

Beyond the economic interpretation, our modeling can also be situated within the wider econometric literature. Stochastic frontier analysis (SFA) has been applied to Indonesian manufacturing to estimate total-factor productivity growth by Ikhsan [6] and to firm-level efficiency more generally Prabowo et al. [19]. Other approaches include generalized method of moments (GMM) and panel cointegration techniques, such as Johansen–Fisher tests, used to study economic growth and financial development in emerging economies [20]. Compared with these methods, the Cobb–Douglas specification with skew-symmetric errors offers two advantages: it retains interpretability of asset and labor elasticities, and it explicitly accounts for asymmetric residual distributions common in emerging markets. Thus, the econometric contribution of this study lies in combining a transparent, interpretable production form with an empirically appropriate error structure for Indonesian firms.

F. Limitations and further suggestions

This study faces two main limitations. First, the dataset is restricted to one to two years of quarterly financial statements from Yahoo Finance, limiting temporal robustness compared to stock return data. Second, our conservative modeling relies on a simplified regression with basic variables, in contrast to studies like Jimichi et al. that incorporate richer firm characteristics. While this approach highlights “pure inputs” and “pure output,” it omits variables such as profits and financial ratios. Future work could extend the dataset and adopt a fuller model of firm financial statements to enhance robustness.

CONCLUSION

This study applies the Cobb–Douglas production function and a skew-symmetric refinement to Indonesian listed firms using quarterly data. The results show that assets exert a stronger influence on revenue than employees, reflecting Indonesia's capital-intensive structure, while the skew-symmetric model improves fit and accounts for asymmetric residuals. Compared with studies in Japan and other markets, our findings highlight that production elasticities vary across economies depending on firm composition and capital diffusion. These results provide new insights into Indonesia's firm-level dynamics and demonstrate

the importance of tailoring econometric specifications to market-specific conditions.

ACKNOWLEDGMENT

Contributor roles in this paper: Conceptualization: G.A.D.N., A.B.; Methodology: G.A.D.N.; Software: G.A.D.N.; Validation: G.A.D.N.; Formal analysis: G.A.D.N.; Investigation: G.A.D.N., A.B.; Resources: G.A.D.N.; Data curation: G.A.D.N.; Writing – original draft: G.A.D.N.; Writing – review and editing: S.S., A.B.; Visualization: G.A.D.N.; Supervision: S.S.; Project administration: G.A.D.N.; Funding acquisition: N/A.

REFERENCES

- [1] Brooks, C. (2014). *Introductory econometrics for finance* (4th ed.). Cambridge University Press. (p. xxix).
- [2] Damodaran, A. (2012). *Investment valuation: Tools and techniques for determining the value of any asset*. John Wiley & Sons. (p. 36).
- [3] Coenders, G., & Serrat, N. A. (2023). Accounting statement analysis at industry level. A gentle introduction to the compositional approach. *arXiv preprint arXiv:2305.16842*.
- [4] Bank Indonesia. (2021). Financial stability review: Safeguarding economic recovery amid financial sector resilience. Bank Indonesia.
- [5] Solow, R. M. (1957). Technical change and the aggregate production function. *Review of Economics and Statistics*, 39(3), 312-320.
- [6] Ikhsan, M. (2007). Total factor productivity growth in Indonesian manufacturing: a stochastic frontier approach. *Global Economic Review*, 36(4), 321-342.
- [7] Azzalini, A. (1985). A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics*.
- [8] Sarkar, J., Goswami, B., Saha, S., & Kar, S. (2018). CD-SFA: stochastic frontier analysis approach to revenue modelling in large cloud data centers. *International Journal of Communication Networks and Distributed Systems*, 21(3), 315-345.
- [9] Jimichi, M., Kawasaki, Y., Miyamoto, D., Saka, C., & Nagata, S. (2023). Statistical Modelling of Financial Data with Skew-Symmetric Error Distributions. *Symmetry*, 15(9), 1772.
- [10] Cobb, C. W., & Douglas, P. H. (1928). A theory of production. *American Economic Review*, 18(1), 139-165.
- [11] Hall, R. E., & Jones, C. I. (1999). Why do some countries produce so much more output per worker than others? *Quarterly Journal of Economics*, 114(1), 83-116.
- [12] Coase, R. H. (1937). The nature of the firm. *Economica*, 4(16), 386-405.
- [13] Demsetz, H. (1973). Industry structure, market rivalry, and public policy. *Journal of Law and Economics*, 16(1), 1-9.
- [14] Coelli, T. J., Rao, D. S. P., O'donnell, C. J., & Battese, G. E. (2005). *An introduction to efficiency and productivity analysis*. Springer science & business media.
- [15] Christensen, L. R., Jorgenson, D. W., & Lau, L. J. (1973). Transcendental logarithmic production frontiers. *Review of Economics and Statistics*, 55(1), 28-45.
- [16] Arellano, M., & Bover, O. (1995). Another look at the instrumental variable estimation of error-components models. *Journal of Econometrics*, 68(1), 29-51.
- [17] International Monetary Fund. (2010). Japan's quest for growth: Exploring the role of capital and innovation (Working Paper No. 10/294). IMF. Retrieved from
- [18] Rudolf, J. (2025, August 21). *Why a commodities boom is not lifting Indonesia's economy*. Financial Times. Retrieved from
- [19] Prabowo, H. E. T., & Cabanda, E. (2011). Stochastic frontier analysis of Indonesian firm efficiency: A note. *International Journal of Banking and Finance*, 8(2), 74-91.
- [20] Gültekin, M., & Umutlu, M. (2023). Economic Growth and Financial Development: Evidence from Panel Cointegration Tests in Emerging Countries. *International Journal of Contemporary Economics and Administrative Sciences*, 13(1), 101-1